DUFITIMANA ARSENE JOHNKELLY

Full Stack Software Engineer | AI & Cloud Systems Developer

Available for High-Impact Roles

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github.com/ArseneBeats Portfolio Website

PROFESSIONAL SUMMARY

Results-driven **Full Stack Software Engineer** and **AI Systems Developer** with proven expertise in designing, building, and deploying scalable web applications, distributed backend microservices, and autonomous AI agents. Highly proficient across modern Frontend (**React.js, Next.js, TypeScript, Tailwind CSS**), distributed Backend architectures (**Python/Django, FastAPI, PHP/Laravel, Node.js**), real-time protocols (**REST, gRPC, WebSockets**), and containerized deployments (**Docker**). Experienced in architecting enterprise solutions ranging from real-time tourism/hotel management ecosystems to intelligent voice-driven AI assistants with RAG pipelines. Bridges software engineering precision with strong backgrounds in cloud infrastructure, IT network systems, and tender/proposal management.

TECHNICAL SKILLS MATRIX

FRONTEND ENGINEERING

React.js | Next.js | TypeScript | JavaScript (ES6+) | Tailwind CSS | Redux Toolkit | Context API | Bootstrap 5 | HTML5 / CSS3 | WebSockets Client | Responsive UI/UX

BACKEND & MICROSERVICES

Python (Django / DRF) | FastAPI | PHP (Laravel) | Node.js / Express | RESTful APIs | gRPC (Protobuf) | WebSockets (Channels) | MVC Architecture | Celery Asynchronous Tasks | C / C++

ARTIFICIAL INTELLIGENCE & LLMS

Autonomous AI Agents (Jarvis AI) | OpenAI API | Google Gemini API | LangChain | RAG Pipelines | Prompt Engineering | Tool Calling / Function Calling | Whisper Speech-to-Text | Neural Text-to-Speech (TTS) | Vector Embeddings

DATABASES, CLOUD & DEVOPS

Docker & Compose | PostgreSQL | MySQL | Oracle DB | Redis (Cache/PubSub) | Git / GitHub Actions (CI/CD) | Linux (Ubuntu Server) | Nginx / Apache | Postman | AWS Fundamentals

NETWORKING, INFRASTRUCTURE & SYSTEMS

TCP/IP & OSI Model | Router & Switch Configuration | LAN/WAN Diagnostics | IT Hardware & Software Maintenance | Cybersecurity Fundamentals | MS 365 Enterprise | Tender / Bid Management | Technical Leadership

PROFESSIONAL EXPERIENCE & KEY PROJECTS

East African TTS (eastafricantts.com) — Lead Full Stack Developer

Live: eastafricantts.com

Project: Tourism, Booking & Hotel Management System

`Django • React.js • REST APIs • Docker • WebSockets • gRPC • PostgreSQL • Redis`

- **Platform Architecture:** Architected and built a comprehensive commercial web application converging regional tour bookings, vehicle rental logistics, and hotel room reservations into a unified digital portal.
- **Real-Time & Distributed Protocols:** Implemented gRPC microservice communication for high-speed inter-service synchronization alongside WebSocket channels for instant booking confirmations and real-time room availability.
- **APIs & Infrastructure:** Engineered secure RESTful APIs with role-based access control, dynamic seasonal pricing, and containerized the frontend, backend, and Redis cache with Docker.

Jarvis AI Assistant — AI Systems Engineer & Full Stack Developer

Personal & Enterprise AI

Project: Autonomous Voice & Conversational Assistant

`Python • FastAPI • Django • React.js • WebSockets • OpenAI / Gemini API • LangChain • Whisper STT • TTS • Docker`

- **Autonomous Agent Workflows:** Architected an intelligent autonomous assistant featuring real-time bi-directional streaming over WebSockets, voice input/output synthesis, and multi-turn conversational reasoning.
- **RAG & Tool Calling:** Integrated custom LangChain RAG pipelines with vector database embeddings for context-aware knowledge queries, combined with function-calling capabilities for OS automation, file operations, and calendar management.
- **Containerization:** Packaged the full backend services and background worker queues into Docker containers for reproducible local and cloud deployment.

Ino Discovery Ltd — Backend Developer — Kigali, Rwanda

Project: Ino Discovery Car sales, rental and Booking System

`Django • React.js • REST APIs • PostgreSQL • Tailwind CSS`

- **Unified Service Platform:** Developed a scalable web application using React.js and Django converging tourism, automotive fleet logistics, and event hosting into a unified ecosystem.
- **Diverse Functionality:** Built distinct modules to handle travel bookings, facilitate car rentals and sales, and manage event logistics with real-time data synchronization.
- **Backend Architecture:** Engineered a robust PostgreSQL database structure supporting complex transactions, vehicle schedules, and real-time ticketing reservations.

J & A Furniture — Backend Developer — E-Commerce System

Project: J&A Furniture E-commerce store

`React.js • Django REST Framework • PostgreSQL • Redux • JWT`

- **Full-Stack Architecture:** Developed a scalable e-commerce platform using React.js for dynamic, responsive frontend interactions and Django REST Framework for backend APIs.
- **Core E-Commerce Functionality:** Implemented product catalog filtering, secure user authentication, shopping cart persistence, and order fulfillment dashboards.
- **User Experience:** Designed an intuitive interface to streamline the customer journey from product discovery to checkout, optimizing performance for both desktop and mobile users.

NIMEN GENERAL BUSINESS (NGB Ltd) — Bid Manager — Kigali, Rwanda

Project: Tender and Proposal Management

`Tender Operations • Public Procurement • Financial Bids • Proposal Architecture • Compliance`

- **Bid Preparation & Submission:** Prepared and submitted professional technical and financial bids and proposals in response to tender opportunities, ensuring 100% compliance with client specifications.
- **Documentation & Accuracy:** Collected, reviewed, and organized technical and financial documents to guarantee accuracy, completeness, and adherence to strict submission deadlines.
- **Cross-Functional Coordination:** Collaborated with technical and management teams to gather project specifications and reviewed past bids to continuously improve proposal success rates.

NIMEN GENERAL BUSINESS (NGB Ltd) IT Support Specialist

Full Time Job

Project: Systems Administration & Hardware Maintenance

`Hardware Maintenance • OS Deployment • Network Diagnostics • Data Backup • Office 365`

- **Technical Support & Maintenance:** Delivered hardware and software maintenance for office computers, printers, and peripheral equipment, ensuring system stability and high availability.
- **Software & Security:** Installed and configured software ensuring compatibility and security; implemented routine data backups and basic recovery support.
- **Network Health:** Conducted routine network monitoring, diagnostics, and resolved workstation connectivity issues across the office.

NIMEN GENERAL BUSINESS (NGB Ltd) Front Desk Clerk

Kigali, Rwanda

Project: Customer Service and Administrative Support

`Client Interaction • Administrative Scheduling • Document Archival • Office Coordination`

- **Client Interaction:** Welcomed clients and visitors, provided prompt information, and directed them to appropriate departments, delivering a professional first impression.
- **Administrative Support:** Managed appointments, handled incoming phone and email inquiries, and maintained accurate records of customer interactions.
- **Document Handling:** Received, reviewed, and organized official corporate documents, ensuring accuracy, structured archiving, and strict confidentiality.

WMTSS Frontend Developer Intern

Kigali, Rwanda

Project: School Management System

`Python • Django • React.js • MySQL • Relational DB Design`

- **Administrative Automation:** Designed and implemented a comprehensive School Management System to digitize manual administrative processes, including student registration and staff scheduling.
- **Core Academic Modules:** Developed modules for tracking daily student attendance, managing grades, and generating automated academic performance report cards.
- **Database Optimization:** Engineered relational database models to securely store and retrieve academic records for hundreds of students with fast query performance.

M & E Company Ltd | Backend and Frontend Developer

Inventory Platform

Project: M & E Stock Management System

`PHP • MySQL • HTML • CSS • JavaScript • Bootstrap`

- **Inventory Control Platform:** Built a robust stock management system designed for real-time inventory tracking, supply chain monitoring, and automated low-stock notifications.
- **Sales & Purchase Ledgers:** Engineered transaction recording for sales and purchase orders, producing automated financial and operational reporting.

WMTSS Club | Backend and Frontend Developer

Retail Systems

Project: B-MIS (Boutique Management Information System)

`PHP • JavaScript • MySQL • POS Architecture`

- **Retail System Development:** Developed and maintained B-MIS for boutique retail operations, handling point-of-sale transactions and inventory management.
- **Code Reviews & Mentorship:** Conducted code reviews and mentored junior developers on modular system design and clean web coding practices.

Pixel Inc Ltd | Frontend Developer Intern

Web Agency

Project: Ecommerce Website Development

`HTML5 • CSS3 • JavaScript • Responsive Design • System Analysis`

- **Responsive Web Development:** Developed eCommerce web pages implementing mobile-first and responsive design principles.
- **UI/UX & Analysis:** Served as web designer and system analyst to optimize user experience, navigation flows, and cross-browser rendering.

IIC | IT Technician

Network Infrastructure

Project: Network Installation & Configuration

`Cisco/MikroTik Routers • Managed Switches • LAN Cabling • IP Addressing • Connectivity`

- **Network Deployment:** Configured local area networks (LAN) using enterprise routers, switches, and patch panels.
- **IP Addressing & Diagnostics:** Implemented IP addressing schemes, subnetting, connectivity testing, and ongoing network troubleshooting and monitoring.

EDUCATION

Bachelor of Science in Software Engineering

Adventist University of Central Africa (AUCA)

Expected Completion: 2027

Software Development Diploma
World Mission High School (Korean School)
2020 – 2023

Computer System Technology & Networking
APAPER Complex School
2018 – 2019

CERTIFICATIONS

- ✓ **AI (Gemini Academy)** — Google Gemini
- ✓ **Full Stack Web Development** — Simplilearn
- ✓ **Advanced SQL Certification** — HackerRank
- ✓ **Oracle DB Certification** — Oracle Dev GYM
- ✓ **Intro to IT & Cybersecurity** — Cybrary
- ✓ **Data Validation Project** — TWB

LANGUAGES

Proficiencies: English (Fluent) • Kinyarwanda (Native) • Korean (Basic) • Kiswahili (Basic)